

Hybrid Solar-Wind-Battery Microgrid Optimization Using Reinforcement Learning for Autonomous Energy Management

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ABSTRACT

The fast convergence of renewable energy has resulted in the emergence of hybrid microgrid systems that are capable of enhancing the sustainability and reliability of energy. Nevertheless, the intermittency and volatility of solar and wind energy sources create serious difficulties to effective energy management and stability of the grid. In this work, the researcher suggests an intelligent optimization system of hybrid solar-wind-battery micro-grids with the help of reinforcement learning to provide an independent approach to energy management as well as enhance the performance of the system. The proposed framework represents the microgrid as a dynamic decision-making system where a reinforcement learning agent acquires the best strategies of energy generation coordination, battery storage scheduling as well as load supply balancing. The system state space comprises the levels of renewable energy production, the state of charge of a battery, and real time loads demand, whereas the action space relates to the energy distribution and storage control options. One of the reward functions is developed to optimise the usage of the renewable energy, reduce energy loss and stabilise the system. The effectiveness of the proposed approach was tested to determine the impact of different conditions of solar irradiance, wind speed, and load demand on simulation experiments. The findings show that the reinforcement learning-based optimization will greatly optimise energy use, increase battery control, and decrease power imbalance when compared to the traditional rule-based control techniques. This suggested intelligent energy management system offers a scalable and adaptive model of hybrid renewable microgrids and can add to the creation of self-organising and robust energy systems in the future smart grid structures.

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1. INTRODUCTION

The world is now in greater need of sustainable and reliable energy systems as a result of the rapid industrialization, population and growing concerns on the environment in relation to traditional power generation systems which relies mainly on fossil fuels. The increasing effects of climatic change and greenhouse gas emissions have led to the acceleration of adoption of renewable energy sources that include solar and wind energy. These are renewable resources whose energy solutions are green and alternative to nonrenewable sources of energy that can significantly lower carbon emission and reliance in fossil fuels [1], [2]. Nevertheless, the nature of renewable energy production in terms of its

intermittency and unpredictability presents significant difficulties in the process of ensuring the stability of power supply and successful grid performance. Hybrid renewable energy systems have been developed as a way out in solving the variability and uncertainty nature of the individual renewable resources. The hybrid systems would enhance the reliability of the systems and allow them to have a continuous power supply through the integration of various renewable energy technologies. Solar-wind microgrids with battery energy storage systems are among other forms of hybrids that have been particularly popular because they can counter the fluctuating generation and demand variability in distributed energy network [3], [4]. Battery storage systems are very significant in such hybrid system as

they store the extra renewable energy at times of maximum generation and provide the stored amounts of energy at the low generation times. Having proper energy management is key to an efficient functioning of the hybrid microgrids. The classical energy management approaches traditionally depend on the rule-based control or the deterministic optimization approach. In as much as these methods are able to give rudimentary control of the operations, they are usually preset to certain operational conditions and lack the flexibility to deal with the changes in environmental parameters like sun irradiance, wind velocity and the load requirement which are dynamic [5]. These constraints get serious as the renewable penetration gets higher and this requires intelligent and adaptive levels of control to be developed. The current superior efficiency of artificial intelligence and machine learning has allowed building intelligent energy management systems that optimise the operations of microgrids under dynamic conditions. Reinforcement learning has become one of the techniques that have proved effective in resolving the sequential decision making problem in complex environments. Reinforcement learning is an approach that enables an intelligent agent to acquire the best control behaviour in the system environment via the constant interaction with the environment through maximising the cumulative rewards [6]. This is what renders reinforcement learning especially convenient in managing energy in renewable microgrids whose conditions in the system change constantly. Some studies have been done on the reinforcement learning to optimise energy in smart grids and renewable energy systems. These methods have been shown to have better load balancing, demand response, and battery scheduling than the traditional optimization methods [7]. Nevertheless, most of the available literature is based on simplified models of systems or individual renewable resources and does not provide a representation of how far they would be relevant to actual hybrid microgrid models. Moreover, it has not been studied extensively by incorporating reinforcement learning into hybrid solar-wind-battery microgrid systems as autonomous energy resources. To overcome these shortcomings, this paper recommends a reinforcement learning-based optimization model to hybrid solar-wind-battery microgrids to help it reach the goal of autonomous energy management. The suggested system will allow the smart coordination of renewable energy sources with battery storage units to optimise the distribution of energy, as well as maximise the use of renewable

energy and stability of the system upon different environmental conditions. The key findings of the research are:

- Hybrid development of renewable model micro grids that combine solar photovoltaic systems, wind turbines and battery energy storage unit.
- Architecture of a reinforcement learning-based energy management system of autonomous microgrids.
- Energy distribution optimization to achieve energy utilisation of renewable energy, as well as balance between power generation.
- Simulation experiment of the proposed system with different environments.

2. RELATED WORK

The benefits that include the ability to result in increased energy reliability, sustainability, and operational efficiency in the distributed power systems have made hybrid renewable energy system a subject of great research attention. Solar photovoltaic and wind energy systems are highly embraced among the renewable source due to their nature of benefiting the environment and the growing technological maturity [8]. The variable characteristics of solar irradiance and wind speeds, however, present some significant difficulties in keeping power production and effective use of energy in the microgrid systems within a stable state [9], [10]. In order to solve these problems, a number of studies have examined hybrid setups of micro grids where solar photovoltaic systems, wind turbines and battery energy storage unit are incorporated. Hybrid renewable systems have the capacity of enhancing the reliability of energy supply by compensating the variation of individual renewable sources. An example is that, at a given time, the solar production is generally large during the day time whereas the wind energy can be produced at other times, thus making it a plan to have supplementary energy production [11]. Battery energy storage systems also increase stability of the systems by storing excess renewable energy when there is high generation and releasing stored energy when there is low generation conditions [12]. The traditional methods of managing energy in hybrid microgrids are usually based on rule-of-thumb control procedures or deterministic control methods. The rule-based approaches are easy to deploy however they do not provide flexibility in a dynamic environment. Micro grid energy management has also highly adopted mathematical optimization

techniques like linear programming, mixed-integer programming and non-mathematical optimization techniques in the form of discretionary algorithms. Although such techniques may offer the best scheduling practises in certain circumstances, they usually demand precise forecasting frameworks and set operational restrictions, which restricts their performance in extremely uncertain renewable energy settings [13]. As the technologies of artificial intelligence evolve fast, the researchers have considered methods of machine learning and deep learning to enhance energy management of a microgrid. Algorithms of machine learning have been used in renewable energy prediction, load demand prediction, and optimising energy dispatched. Models of deep learning have shown promising images of modelling nonlinear varied relationships among environmental variables and energy production patterns [14]. Reinforcement learning has become one of the potent approaches to solving sequential decision problems in dynamic environments due to its AI-based solving and application. The reinforcement learning is used to help an intelligent agent to learn optimal control policies through interaction with the environment and maximising cumulative rewards over time. Reinforcement learning in energy systems was used in several applications such as smart grid control, optimization of demand response, scheduling of batteries and management of distributed energy resources [15]. A number of recent researches have indicated reinforcement learning effectiveness in micro grid energy management. As one instance, controllers based on reinforcement learning have been applied to optimization of battery charge-discharge scheduling and minimization of operational costs in renewable microgrids [16]. The deep reinforcement learning methods have been investigated in other studies to manage demand response and energy trading in smart grids [17]. These techniques have demonstrated enhanced flexibility and functional efficiency over the conventional optimization techniques.

Since this has been achieved, there are still a few limitations in the current literature. Most of the research is on simplified system structures or individual renewable energy sources, which puts a restriction on its applicability to hybrid microgrid real-world settings. Moreover, the design of reinforcement learning to be implemented in combination with hybrid solar-wind-battery systems to autonomously control microgrids has not been completely investigated. Issues pertaining to reward

design function, scalability of system, and real-time decision-making are research questions that are still open. To overcome these shortcomings, this paper will introduce a reinforcement learning-based optimization model of a hybrid solar-wind-battery microgrids that will allow natural management of energy in a changing environment. The framework proposed will combine the formulation of renewable energy, including battery storage and control as well as block of decision to increase usage of renewable energy and improve stability of the systems in distributed micro grids.

3. HYBRID MICROGRID SYSTEM ARCHITECTURE

The hybrid microgrid system that has been proposed to be introduced into the proposed distributed energy environment combines various renewable energy sources and an energy storage unit to provide stable electricity to local loads. The microgrid architecture comprises of solar photovoltaic generation, wind turbine generation and a battery energy storage system, and a load demand unit which is an example of domestic or commercial power consumption. That being said, these elements are synchronised by an intelligent microgrid controller that makes it possible to control autonomously energy management via reinforcement learning techniques.

3.1 Solar Photovoltaic System Model

The solar photovoltaic (PV) system transforms the solar irradiance into electrical energy with the help of photovoltaic modules. The solar radiation intensity and the efficiency of photovoltaic modules are the major determinants of the power that PV system can produce. The PV system can be described as output power as:

$$P_{PV} = \eta_{PV} \times A \times G \quad (1)$$

where P_{PV} represents the power generated by the photovoltaic array, η_{PV} denotes the efficiency of the photovoltaic modules, A represents the surface area of the solar panels, and G represents the solar irradiance measured in watts per square meter. The changes in PV power generation are as a result of variability in solar irradiance due to sources of environmental variations like cloud cover, and changing seasons.

3.2 Wind Turbine Power Generation Model

Wind turbines turn the power of a wind into electrical energy by employing rotational blades attached to an electrical generator. The output of the wind power turbine is primarily determined by the speed of wind

and the nature of turbines. The theoretical availability of wind power on a turbine is computed using the equation below:

$$P_{wind} = \frac{1}{2} \rho A V^3 C_p \quad (2)$$

where P_{wind} represents the generated wind power, ρ denotes air density, A represents the rotor swept area, V denotes wind speed, and C_p represents the power coefficient of the turbine. Wind energy generation is very heavily weather-dependent since the cubic relationship between power generated and wind speed makes the variation in power generation significantly change.

3.3 Battery Energy Storage System Model

Microgrid balancing the supply and demand of energy is done through the use of battery energy storage system. At these times when the renewable energy production is high, excess energy is stored in the battery, whereas the stored energy is released at times when production of renewable energy is not high enough to absorb the load. A battery state of charge (SOC) is a significant measure that illustrates the quantity of energy that can be availed of the battery. The equation below can be used to model the SOC:

$$SOC_{t+1} = SOC_t + \frac{P_{charge} - P_{discharge}}{C_{battery}} \quad (3)$$

where SOC_t represents the battery state of charge at time t , P_{charge} represents the charging power, $P_{discharge}$ represents the discharging power, and $C_{battery}$ represents the total battery capacity.

The battery has preset operational ranges to eliminate the chance of overcharging and over-discharging that would eventually shorten the battery life.

3.4 Microgrid Control Architecture

The hybrid microgrid is regulated by a central control unit tasked with overseeing the production of energy, the amount of the battery storage facility, and the demand conditions of the load. The controller constantly gathers information about the systems and determines the best strategies of energy distribution. The control architecture has a reinforcement learning agent implemented in it to allow the microgrid operation to be intelligently ruled. The agent deals with the environment of microgrids and optimally learns optimal control policies that strike a balance between energy supply and demand as well as maximising the use of renewable energy. Fig. 1 shows the general design of the proposed hybrid micro grid system.

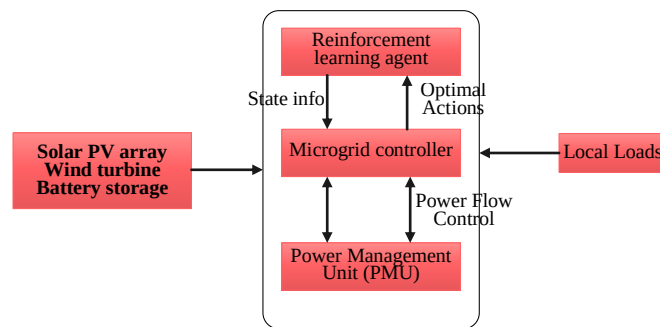


Fig. 1. Hybrid Solar-Wind-Battery Microgrid Architecture

4. REINFORCEMENT LEARNING-BASED ENERGY MANAGEMENT FRAMEWORK

As the fundamental optimization system used to control the hybrid microgrid system, reinforcement learning is used. Reinforcement learning involves an intelligent agent engaging with an environment to acquire optimal policies in a process that is guided by trial-and-error. The objective of the learning process is to optimise the cumulative rewards, acquired by decision-making in series.

4.1 Reinforcement Learning Environment

The reinforcement learning environment is composed of the microgrid system. Environment brings information to the state of the systems such as renewable energy generation levels, the charge in a battery and load demand. On the basis of this, the reinforcement learning agent chooses control actions that define how energy will be distributed among the system components.

4.2 State Space Representation

The state representation explains the present state of the microgrid system. The state variables in the solar power generation, wind power generation, battery state of charge, and the load demand. These parameters enable the reinforcement learning agent to know whether the microgrid is operational or not and make optimized control decisions.

Mathematically this state of the system can be represented as:

$$S_t = (P_{solar}, P_{wind}, SOC, D_{load}) \quad (4)$$

where S_t represents the system state at time step t .

4.3 Action Space

The reinforcement learning agent establishes the distribution of the energy within the microgrid. The potential control measures will be to power the battery by using surplus renewable power, to power the battery by powering it down, and to power the load by electricity of renewable power sources. The choice that the agent makes has a direct impact on the energy balance of microgrids and the output of the system.

4.4 Reward Function

The reward mechanism assesses every action chosen by the learning agent of the reinforcement. The reward function is aimed to motivate the activities designed to advance the use of renewable energy sources, reduce the loss of energy, provide stable power sources, prevent overcharging the batteries, or over-discharge.

The rewarding capability can be determined as:

$$R_t = \alpha U_{renewable} - \beta L_{loss} - \gamma P_{imbalance} \quad (5)$$

where R_t represents the reward at time step t , $U_{renewable}$ represents renewable energy utilization, L_{loss}

represents energy loss within the system, and $P_{imbalance}$ represents power imbalance between generation and demand. The coefficients α , β , and γ represent weighting parameters used to control the importance of each objective.

4.5 Reinforcement Learning Algorithm

The process of reinforcement learning training is that of an iterative learning process where the agent is involved in consecutive interactions with microgrid environment. In every iteration, the agent makes an observation of the existing system state, chooses an action as dictated by its policy, gets a reward of the environment and alters its (agent) learning parameters. The algorithm of the interaction between the microgrid environment and the reinforcement learning agent is presented in Fig. 2. The process of the reinforcement learning applied to the microgrid energy management is summarised in algorithm 1.

Algorithm 1: Reinforcement Learning-Based Energy Management

Step 1: Initialize reinforcement learning agent and environment.

Step 2: Observe the current system state.

Step 3: Select an action according to the current policy.

Step 4: Execute the action in the microgrid environment.

Step 5: Observe the next system state and reward.

Step 6: Update the learning policy using the reward feedback.

Step 7: Repeat the process until convergence is achieved.

With the recurrent encounters with the environment, the agent will learn successful energy management techniques to operate the microgrid effectively.

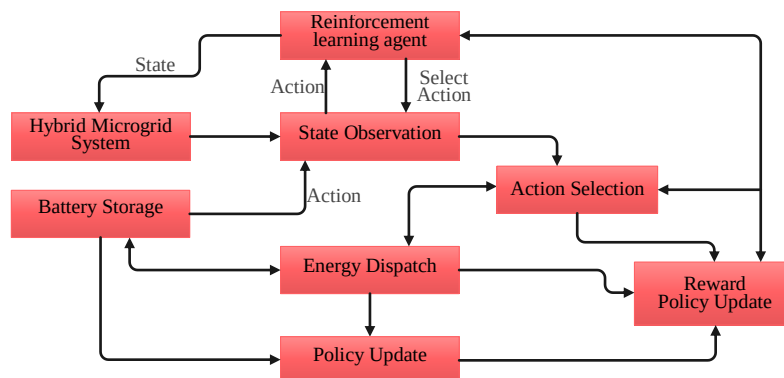


Fig. 2. Reinforcement Learning Energy Management Flowchart

5. SIMULATION SETUP AND RESULTS

5.1 Simulation Setup

The effectiveness of the suggested reinforcement learning-mediated microgrid energy management system was tested with simulation experiments. The simulated environment represents a hybrid microgrid of renewable energy that is comprised of solar photovoltaic panels, wind turbines, battery energy storage units and dynamic load demand. The computational modelling tools that were used to carry out the simulation were able to model dynamics of the renewable energy generation and control strategies of reinforcement learning. Solar irradiance and wind speed were also simulated in an environmental variety to simulate conditions of renewable power generation in real situations. During the simulation, the load demand profile was also fluctuated to reflect fluctuations in the patterns of electricity consumption in the micro grid. The agent by reinforcement learning interacted with the micro grid environment repeatedly to be trained. In each of the training episodes, the agent monitored the existing system state and picked actions managing energy that maximised the distribution of power between solar, wind, and battery elements. This learning process would go on until the agent gets closer to a stable control policy.

5.2 Performance Evaluation

The outcomes of the simulations prove that the reinforcement learning-based control strategy can be successfully used to balance the generation of renewable energy and the load demand in the hybrid microgrid system. The suggested solution enhances the use of renewable energy by smart control of solar and wind generation responsibilities in relation to the battery storage ones. When there is a high level of renewable generation, the energy is stored in the battery system hence the microgrid does not experience any power outage during low periods of generation. The smart control approach also reduces the use of energy wasted and increases battery utilisation effectiveness. The reinforcement learning-based framework has better responses to dynamic environmental conditions and load demand rates when compared to traditional energy management systems based on rules. This is because the reinforcement learning agent would guarantee autonomous and effective operation of the microgrid

due to its continued learning as the system interacts with it.

6. RESULTS AND DISCUSSION

The behaviour of a proposed framework which is based on reinforcement learning energy management was tested by the simulation experiments to cheque the behaviour of the framework under different conditions of renewable generation and load demand. The outcomes of the simulation reveal that the intelligent control strategy is practical in order to balance the generation of renewable energy, the use of battery stored energy, and the load requirement in the hybrid microgrid milieu.

6.1 Renewable Energy Generation Performance

The microgrids in the hybrid system are mainly the solar photovoltaic and wind turbine systems since they offer the main supply of renewable energy. Fluctuations in the sun intensity and the velocity of wind were modelled to reflect the real conditions in the environment. The findings show solar power production is based on the daily irradiance profiles with the greatest power production occurring in the middle of the day. The production of wind energy has stochastic variations in variations of wind speed. This is because when solar energy and wind power are combined, the balance of the renewable power supply increases. The energy shortfall is partially filled by wind power when the output of solar generation declines in the evening hours. The reliability of hybrid micro grid system is augmented with this complementary behaviour.

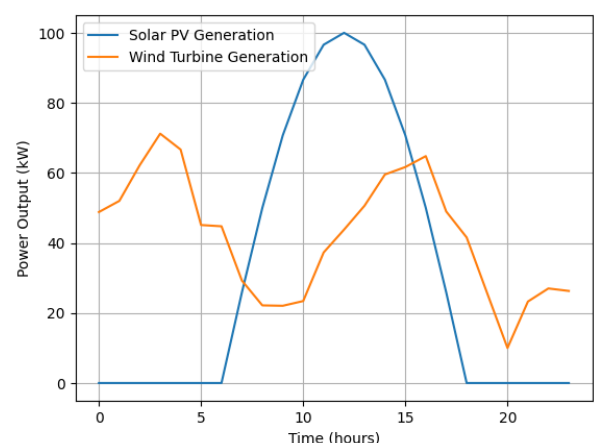


Fig. 3. Solar and wind power generation profiles.

The findings substantiate the fact that hybrid renewable generation enhances the accessibility of

energy and decreases the reliance on one source of renewable energy.

6.2 Battery Energy Storage Performance

The battery energy storage system is very essential in terms of sustaining balance of energy in the micro grid. When there is abundance of renewable energy production, the battery holds up the energy produced and when there is a low end production the battery releases the energy which is used by the load.

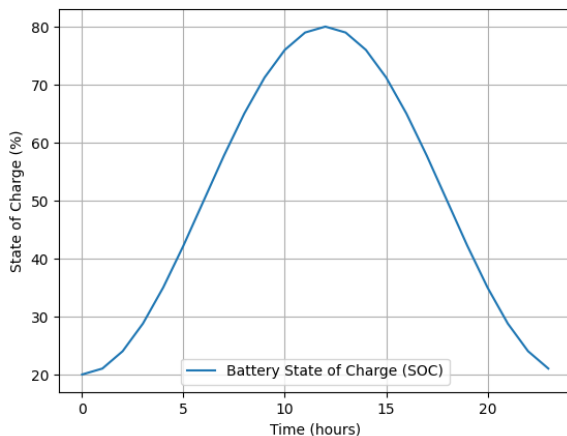


Fig. 4. Battery state-of-charge (SOC) variation during system operation.

Results of simulation indicate that reinforcement learning controller is effective in controlling battery discharging and charging cycles. The SOC is operating inside the safe limits so that neither overcharging nor deep discharge is experienced. Efficient battery scheduling enhances the utilisation of the energy storage and increases the battery life.

6.3 Energy Management Performance

The efficiency of the suggested reinforcement learning-based energy management framework was tested by the comparison to a traditional rule-based control strategy. Measures of performance which included the use of renewable energy, efficiency of the system and energy loss were examined. Table 1 shows the results of the comparative performance.

Table 1. Performance comparison of energy management strategies

Method	Renewable Utilization (%)	Energy Loss (%)	System Efficiency (%)
Rule-Based Control	72	12	85
RL-Based Optimization	89	5	93

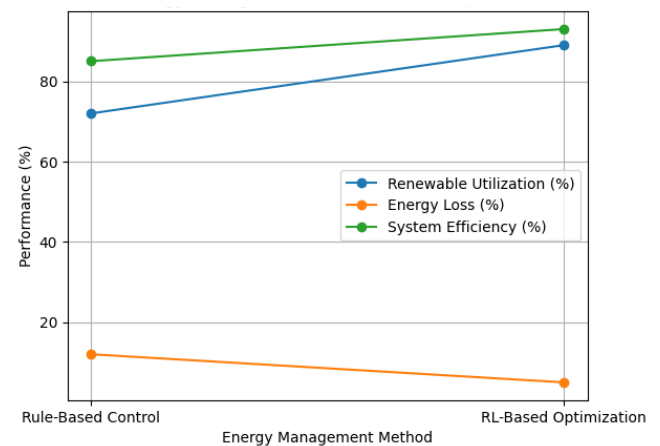


Fig. 5. Energy management performance comparison between rule-based control and reinforcement learning-based optimization.

The findings indicate that the reinforcement learning method is an important way to enhance the use of renewable energy and minimise its losses in the system. The system that is based on the RL algorithm is known to achieve a higher level of renewable energy use (around 17 percent) than the traditional methods of control.

6.4 Adaptive Control Performance

Reinforcement learning agent is an ongoing learning process that relies on the interaction between the agent and the micro grid environment. The dynamic nature of this learning capability allows the system to dynamically tune the energy management decisions according to the fluctuations in the environment conditions and load demand patterns. The adaptive control system enhances the stability of the systems. In contrast to conventional energy management methods which are based on control policies that have been pre-defined, a reinforcement learning agent can discover the optimal policies of control independently by being fed back on by the environment. This is agreeable with other works of literature that have shown the efficiency of reinforcement learning in optimization of energy in smart grids and microgrids. Indicatively, recent studies indicated that strategies of energy management through reinforcement learning can be very effective in enhancing energy efficiency and grid stability over more deterministic optimization methods. All in all, the proposed reinforcement learning-based hybrid microgrid management framework offers better performance of the system, better utilisation of renewable energy and adaptive

capability of control that can be applicable to the next generation of a distributed energy systems.

CONCLUSION

This paper introduced a smart energy management system of hybrid solar-wind-battery microgrids with the use of a reinforcement-based optimization system. The challenges that the intermittency of renewable energy resources faced, alongside the necessity to have effective coordination between energy generation, storage, and load demand in a micro grid distributed environment, were to be dealt with through the proposed system. A hybrid micro grid system, which incorporates solar photovoltaic farming, wind turbine systems and battery energy storage was designed and modelled. An autonomous control measure designed to allow autonomous decision-making so that energy is distributed optimally was introduced by implementing a reinforcement learning control approach. The reinforcement learning agent kept on operating in the microgrid environment and trained optimal policies which maximise the use of renewable energy and at the same time ensure the stability of the system. The outcomes of the simulation evidenced that the suggested framework is an effective way of balancing between renewable generation and load demand and enhancing the use of batteries and minimising energy losses. The reinforcement learning-based approach was better in terms of renewable energy use and the system overall efficiency as compared to the conventional rule-based control strategies. This is because the adaptive learning ability of the reinforcement learning agent enables the microgrid to be dynamically responsive to changes in the environment conditions and load demand without the predetermined control rules. The primary outcomes of the given research are the creation of a renewable microgrid model based on the hybrid regime of solar, wind, and battery storage, the creation of a reinforcement learning-driven autonomous energy management system, and the characterization of the proposed system functionality under diverse operating conditions. The findings show that smart optimization methods, like the reinforcement learning, can greatly improve the work performance and stability of renewable microgrids. Future studies can concentrate on the implementation of deep reinforcement learning algorithms that can promote the performance of decision-making, integration of demand response strategies to achieve better energy

flexibility, and practical testing of the offered framework in real-world conditions of microgrids. Further research on the use of multi-agent reinforcement learning to control coordinated energy management in an interconnected microgrid network can also be explored in the future.

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